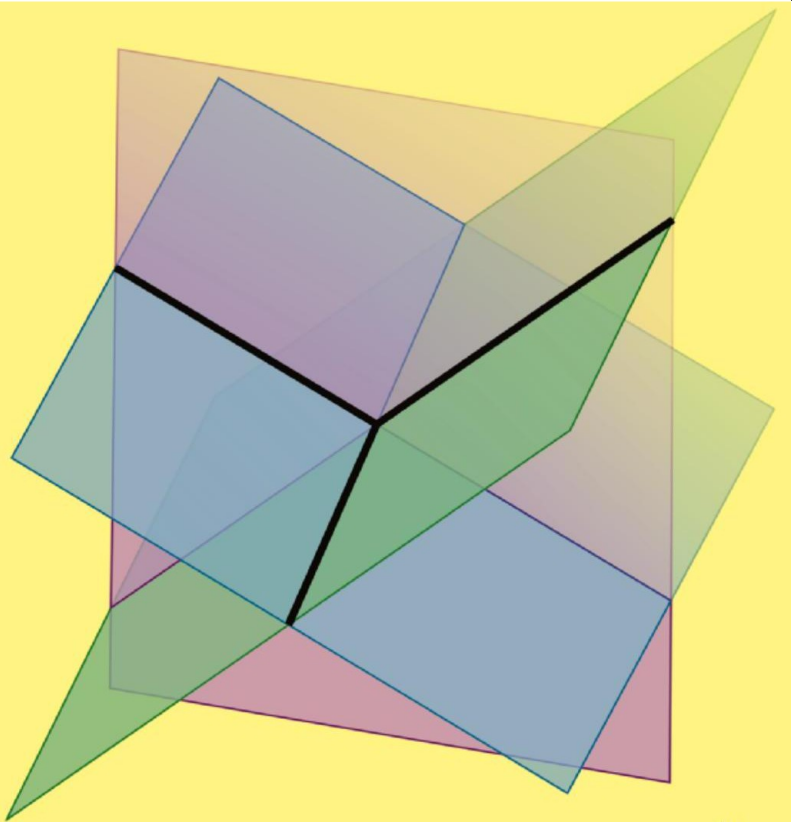
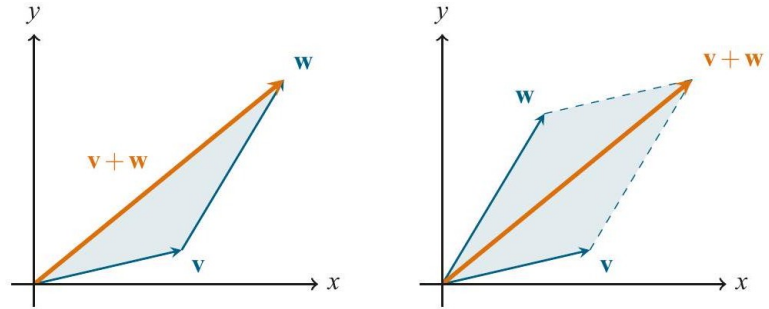
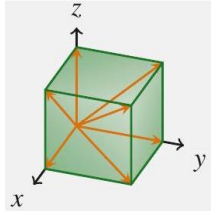
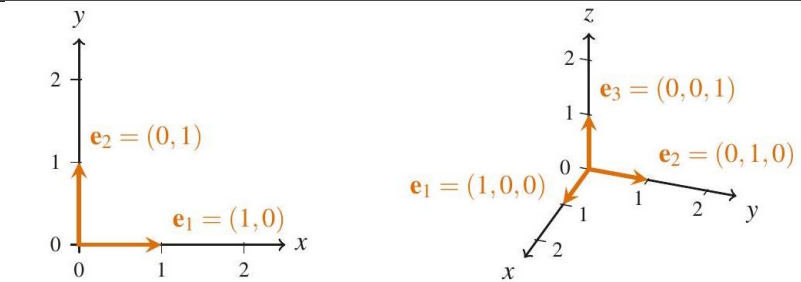
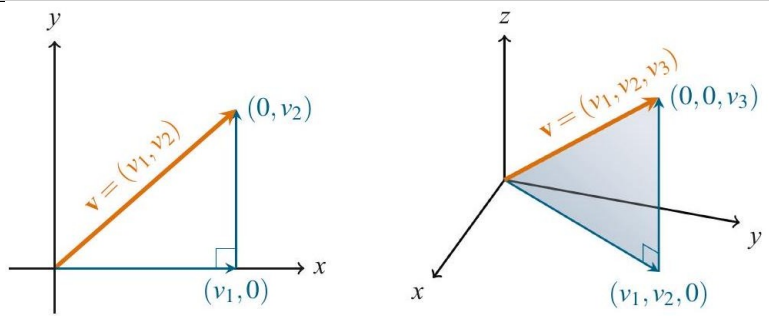
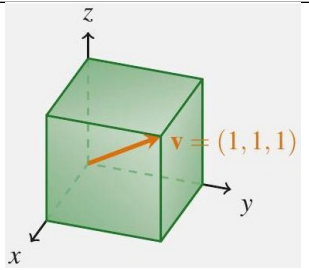
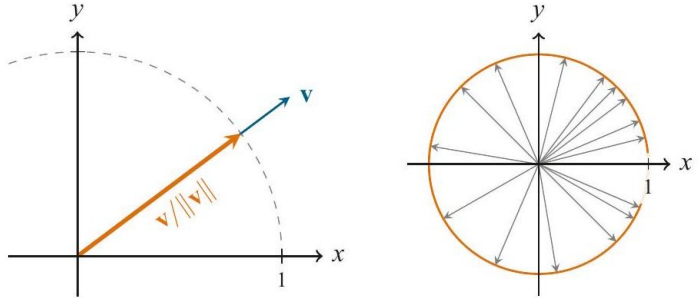
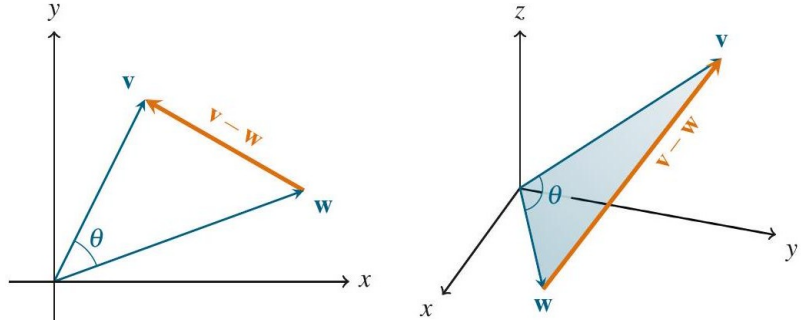
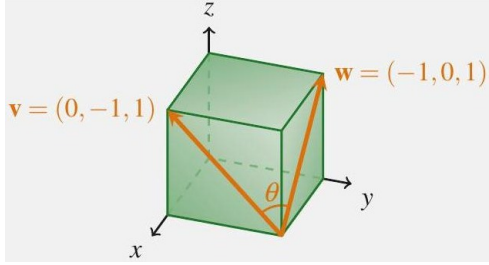
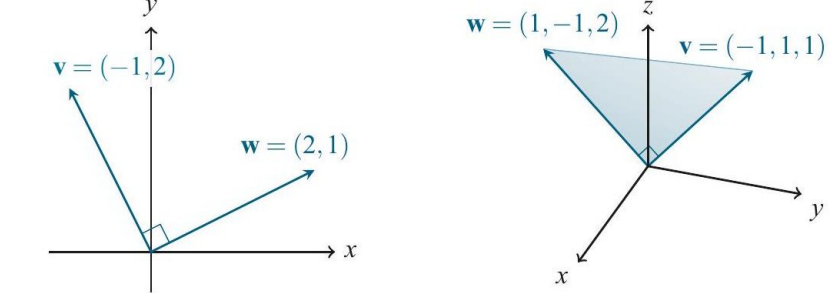
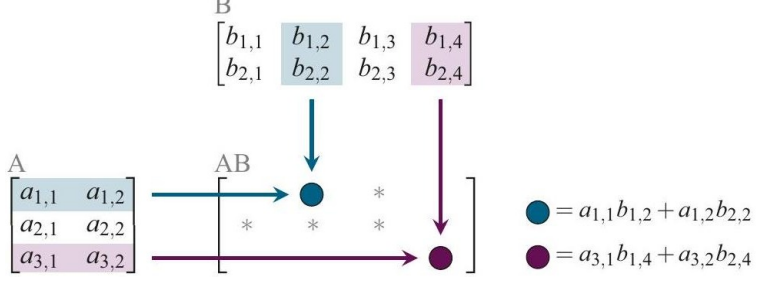


Image regions

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_ DIA_0001	1	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-DIA_0002	16	—	—	—	Chapter 1: Vectors and Geometry Chapter 2: Linear Systems and Subspaces Chapter 3: Unraveling Matrices Advanced Linear and Matrix Algebra
johnston-linear-matrix-algebra-DIA_0003	17	—	—	—	(a) The vector $\mathbf{v} = (3, 2) \in \mathbb{R}^2$. (b) The vector $\mathbf{v} = (1, 3, 2) \in \mathbb{R}^3$.

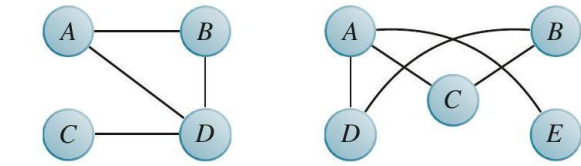
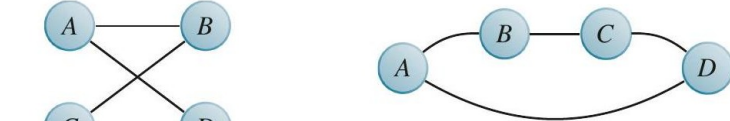
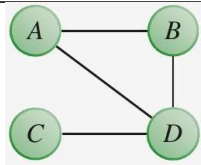
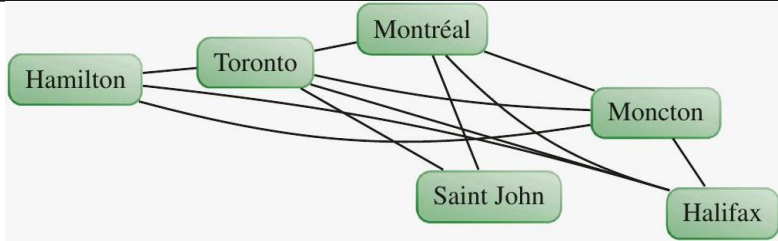
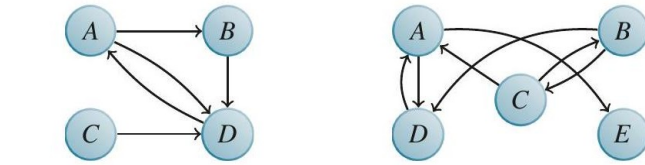
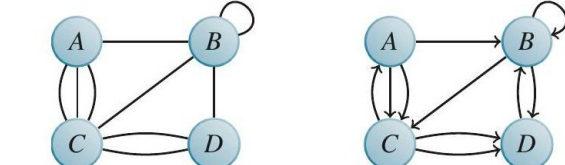
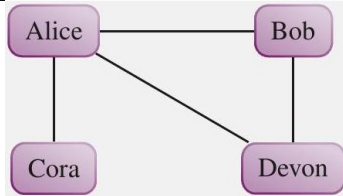
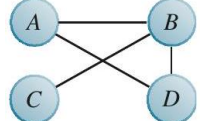
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_DIA_0004	18	—	—	—	 (a) Adding vectors head-to-tail. (b) Adding vectors in standard position.
johnston-linear-matrix-algebra_DIA_0005	19	—	—	—	
johnston-linear-matrix-algebra_DIA_0006	23	—	—	—	 (a) The standard basis vectors e_1 and e_2 in $\mathbb{R}^2$. (b) The standard basis vectors e_1, e_2, and e_3 in $\mathbb{R}^3$.
johnston-linear-matrix-algebra_DIA_0007	27	—	—	—	property (a) $(c\mathbf{v}) \cdot \mathbf{w} = \mathbf{w} \cdot (c\mathbf{v}) = c(\mathbf{w} \cdot \mathbf{v}) = c(\mathbf{v} \cdot \mathbf{w})$. property (c)
johnston-linear-matrix-algebra_DIA_0008	28	—	—	—	 (a) A vector $\mathbf{v} \in \mathbb{R}^2$. (b) A vector $\mathbf{v} \in \mathbb{R}^3$.
johnston-linear-matrix-algebra_DIA_0009	29	—	—	—	

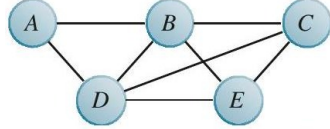
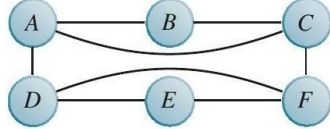
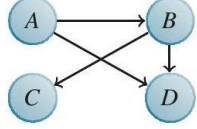
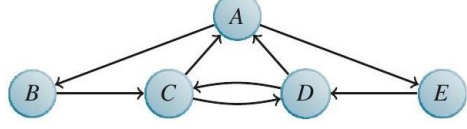
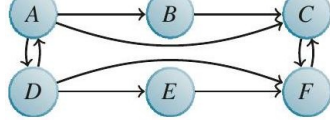
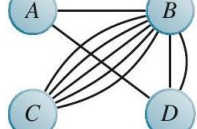
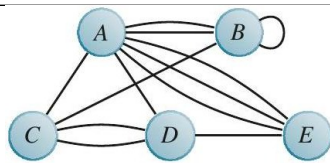
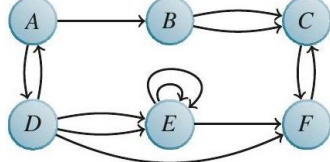
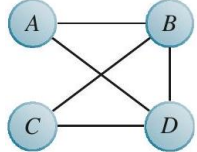
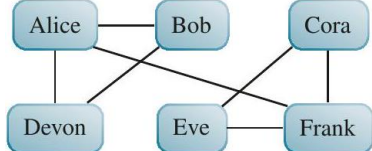
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_DIA_0010	30	—	—	—	 (a) Renormalizing a vector $\mathbf{v} \in \mathbb{R}^2$. (b) The unit circle in $\mathbb{R}^2$.
johnston-linear-matrix-algebra_DIA_0011	32	—	—	—	 (a) Vectors $\mathbf{v}$, $\mathbf{w}$, and $\mathbf{v} - \mathbf{w}$ in $\mathbb{R}^2$. (b) Vectors $\mathbf{v}$, $\mathbf{w}$, and $\mathbf{v} - \mathbf{w}$ in $\mathbb{R}^3$.
johnston-linear-matrix-algebra_DIA_0012	33	—	—	—	
johnston-linear-matrix-algebra_DIA_0013	34	—	—	—	 (a) The vectors $(-1, 2)$ and $(2, 1)$ in $\mathbb{R}^2$ are orthogonal since $(-1, 2) \cdot (2, 1) = -2 + 2 = 0$. (b) The vectors $(-1, 1, 1)$ and $(1, -1, 2)$ in $\mathbb{R}^3$ are orthogonal since $(-1, 1, 1) \cdot (1, -1, 2) = -1 - 1 + 2 = 0$.
johnston-linear-matrix-algebra_DIA_0014	38	—	—	—	 $\bullet = a_{1,1}b_{1,2} + a_{1,2}b_{2,2}$ $\bullet = a_{3,1}b_{1,4} + a_{3,2}b_{2,4}$

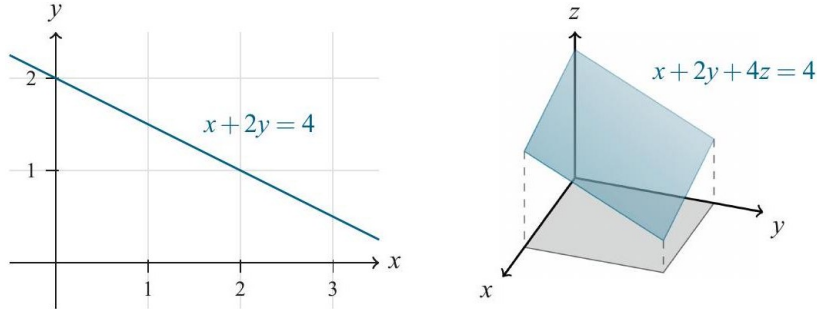
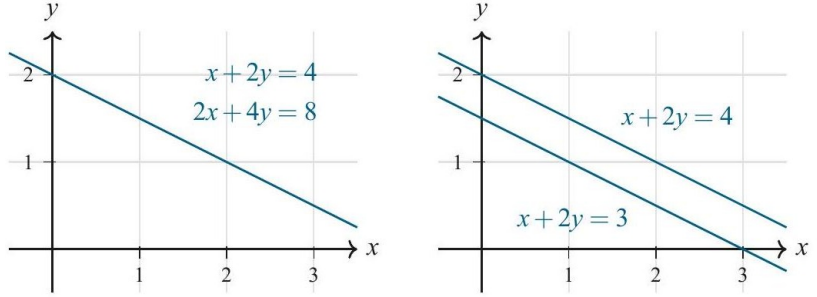
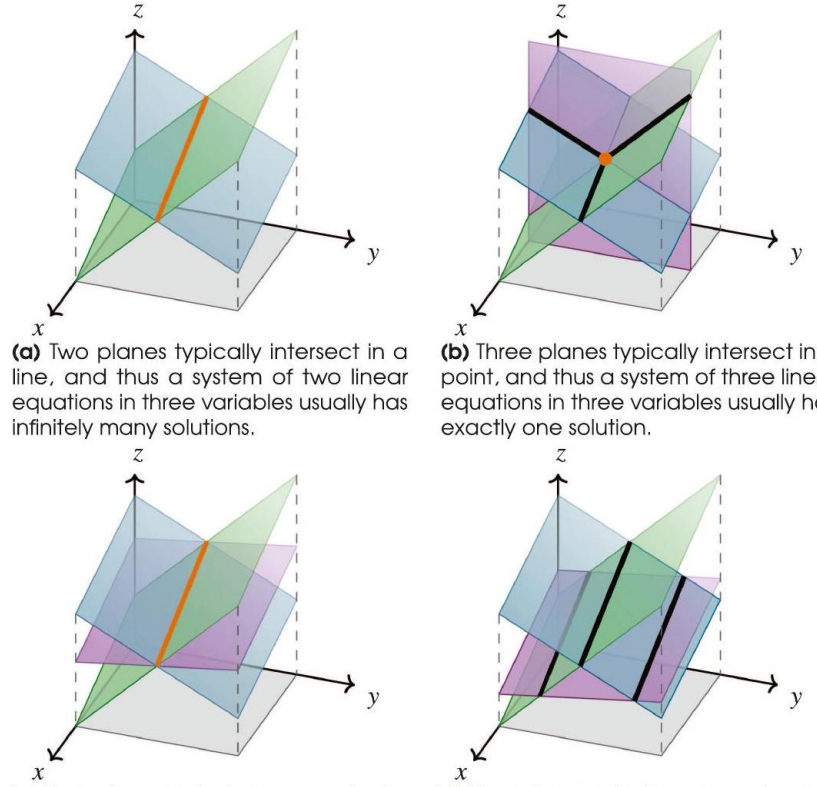
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_ DIA_0015	39	—	—	—	
johnston-linear-matrix-algebra_ DIA_0016	51	—	—	—	
johnston-linear-matrix-algebra_ DIA_0017	57	—	—	—	
johnston-linear-matrix-algebra_ DIA_0018	58	—	—	—	
johnston-linear-matrix-algebra_ DIA_0019	59	—	—	—	
johnston-linear-matrix-algebra_ DIA_0020	59	—	—	—	

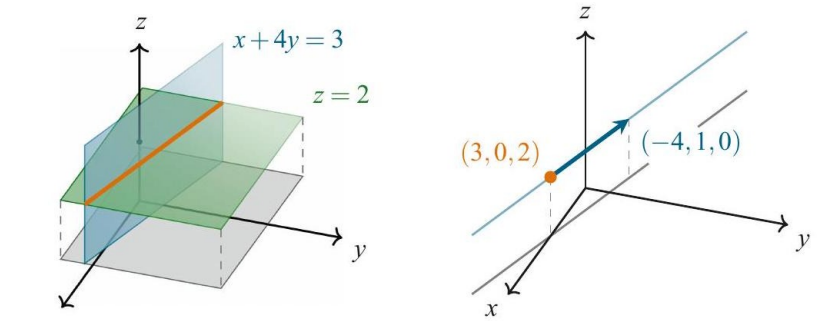
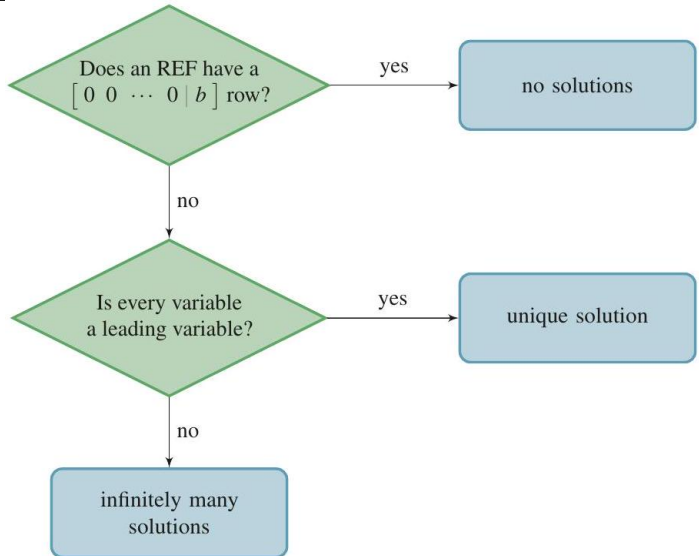
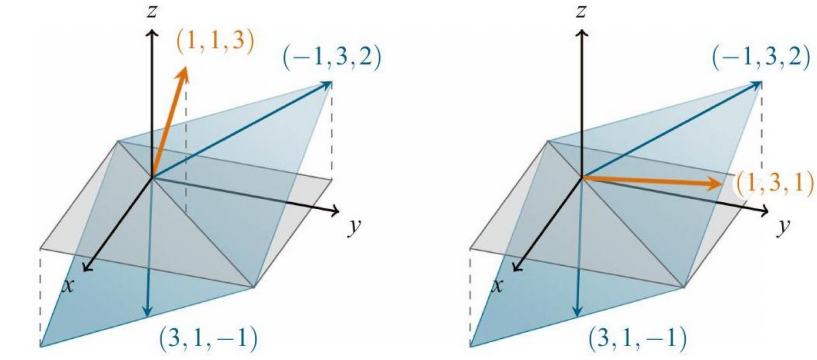
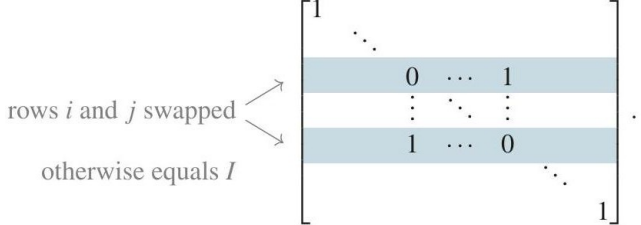
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_DIA_0021	61	—	—	—	
johnston-linear-matrix-algebra_DIA_0022	61	—	—	—	
johnston-linear-matrix-algebra_DIA_0023	62	—	—	—	(a) Rotating $\mathbf{v}$ and $\mathbf{w}$ and then adding them has the same effect as adding and then rotating them, so $R^\theta(\mathbf{v} + \mathbf{w}) = R^\theta(\mathbf{v}) + R^\theta(\mathbf{w})$. (b) Rotating $\mathbf{v}$ and then multiplying it by c has the same effect as multiplying by c and then rotating, so $R^\theta(c\mathbf{v}) = cR^\theta(\mathbf{v})$.
johnston-linear-matrix-algebra_DIA_0024	65	—	—	—	
johnston-linear-matrix-algebra_DIA_0025	66	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_DIA_0026	67	—	—	—	
johnston-linear-matrix-algebra_DIA_0027	74	—	—	—	(a) A vector $\mathbf{w}$ that is orthogonal to a given vector $\mathbf{v} \in \mathbb{R}^2$. (b) The cross product $\mathbf{v} \times \mathbf{w}$ is orthogonal to both $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^3$.
johnston-linear-matrix-algebra_DIA_0028	74	—	—	—	$\mathbf{v} \times \mathbf{w} = (v_2 w_3 - v_3 w_2, v_3 w_1 - v_1 w_3, v_1 w_2 - v_2 w_1)$
johnston-linear-matrix-algebra_DIA_0029	76	—	—	—	(a) A parallelogram with sides $\mathbf{v}$ and $\mathbf{w}$. (b) The triangle formed by $\mathbf{w}$.
johnston-linear-matrix-algebra_DIA_0030	78	—	—	—	height: $\ \mathbf{v}\ \cos(\theta)$ base area: $\ \mathbf{w} \times \mathbf{x}\$

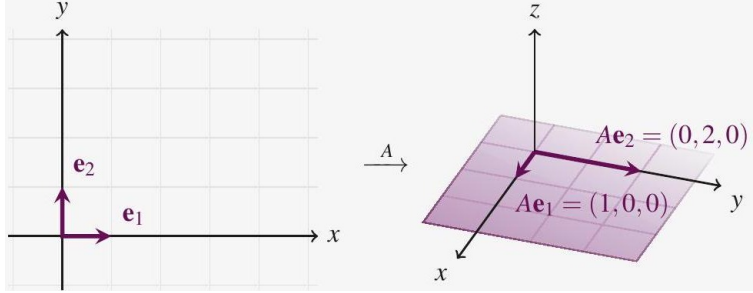
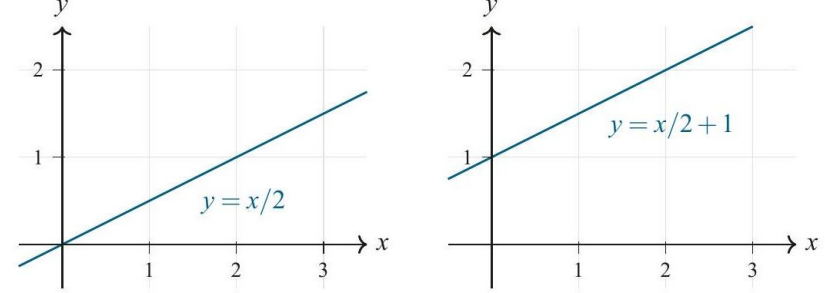
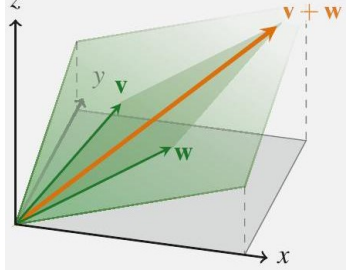
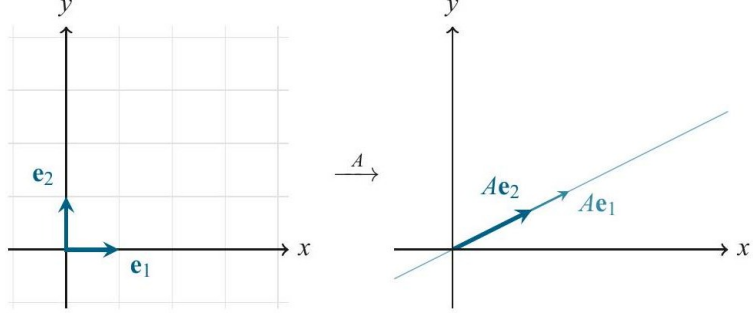
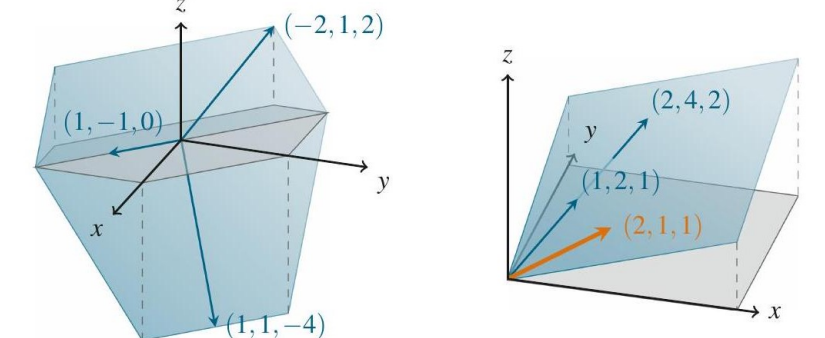
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_DIA_0031	80	—	—	—	 (a) A graph with 4 vertices. (b) A graph with 5 vertices.
johnston-linear-matrix-algebra_DIA_0032	81	—	—	—	 (a) A graph with 4 vertices. (b) The same graph as the one on the left, but represented in a different way.
johnston-linear-matrix-algebra_DIA_0033	83	—	—	—	
johnston-linear-matrix-algebra_DIA_0034	84	—	—	—	
johnston-linear-matrix-algebra_DIA_0035	86	—	—	—	 (a) A directed graph with 4 vertices. (b) A directed graph with 5 vertices.
johnston-linear-matrix-algebra_DIA_0036	87	—	—	—	 (a) An undirected multigraph. (b) A directed multigraph.
johnston-linear-matrix-algebra_DIA_0037	88	—	—	—	
johnston-linear-matrix-algebra_DIA_0038	90	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-DIA_0039	90	—	—	—	
johnston-linear-matrix-algebra-DIA_0040	90	—	—	—	
johnston-linear-matrix-algebra-DIA_0041	90	—	—	—	
johnston-linear-matrix-algebra-DIA_0042	90	—	—	—	
johnston-linear-matrix-algebra-DIA_0043	90	—	—	—	
johnston-linear-matrix-algebra-DIA_0044	90	—	—	—	
johnston-linear-matrix-algebra-DIA_0045	90	—	—	—	
johnston-linear-matrix-algebra-DIA_0046	90	—	—	—	
johnston-linear-matrix-algebra-DIA_0047	90	—	—	—	
johnston-linear-matrix-algebra-DIA_0048	91	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_DIA_0049	93	—	—	—	 (a) A linear equation in 2 variables represents a line in $\mathbb{R}^2$. The equation of this line is perhaps more usually written in the (equivalent) form $y = -x/2 + 2$. (b) A linear equation in 3 variables represents a plane in $\mathbb{R}^3$. This plane actually extends arbitrarily far in all directions—it is just truncated here so it is easier to visualize.
johnston-linear-matrix-algebra_DIA_0050	94	—	—	—	 (a) A linear system with infinitely many solutions, represented by two equivalent (and thus always intersecting) lines. (b) A linear system with no solutions, represented by two parallel (and thus non-intersecting) lines.
johnston-linear-matrix-algebra_DIA_0051	97	—	—	—	 (a) Two planes typically intersect in a line, and thus a system of two linear equations in three variables usually has infinitely many solutions. (b) Three planes typically intersect in a point, and thus a system of three linear equations in three variables usually has exactly one solution. (c) Three (or more) planes can also intersect in a line, and thus a system of three or more linear equations in three variables can have infinitely many solutions. (d) Three (or more) planes can also not have a common intersection at all, and thus a system of three or more linear equations in three variables can have no solutions.

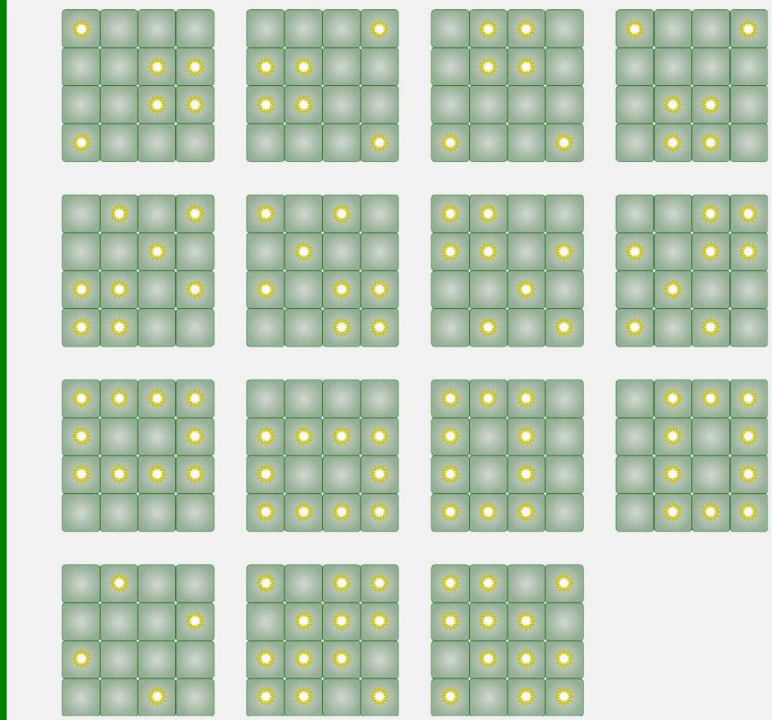
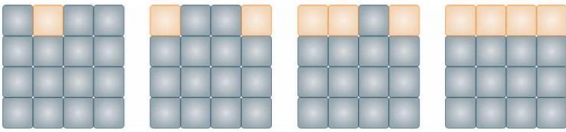
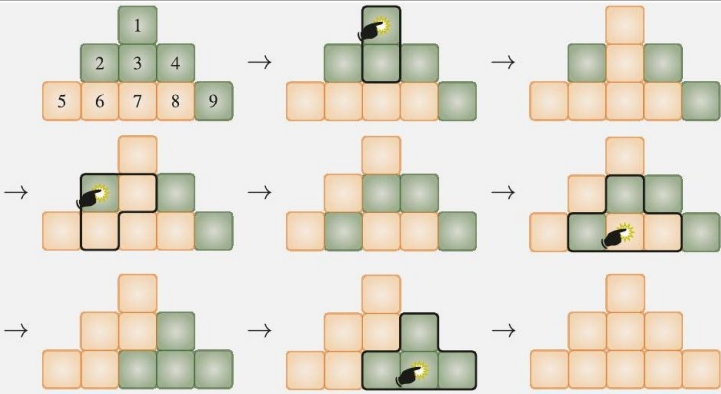
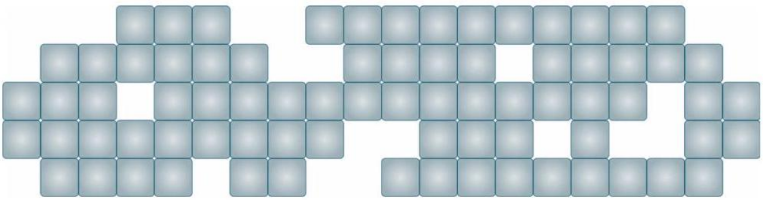
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra- DIA_0052	106	—	—	—	 (a) The solution set is the intersection of the two planes $x + 4y = 3$ and $z = 2$. (b) The solution set is the line going through the point $(3, 0, 2)$ in the direction of the vector $(-4, 1, 0)$.
johnston-linear-matrix-algebra- DIA_0053	106	—	—	—	
johnston-linear-matrix-algebra- DIA_0054	111	—	—	—	 (a) $(1, 1, 3)$ is not a linear combination of $(-1, 3, 2)$ and $(3, 1, -1)$. (b) $(1, 3, 1)$ is a linear combination of $(-1, 3, 2)$ and $(3, 1, -1)$.
johnston-linear-matrix-algebra- DIA_0055	117	—	—	—	

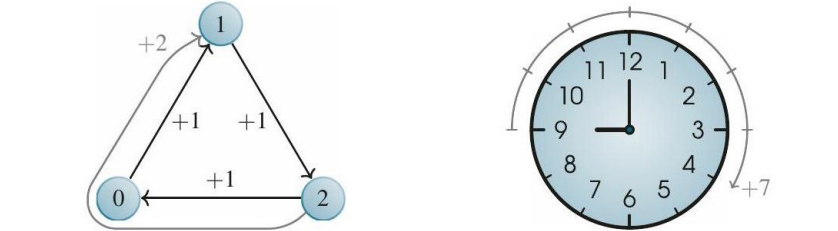
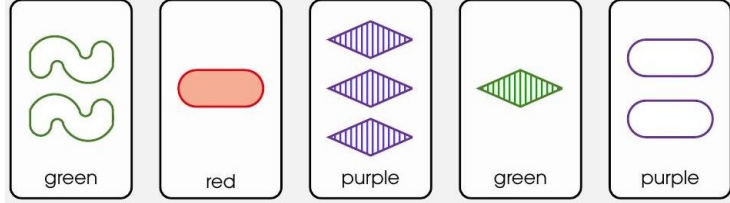
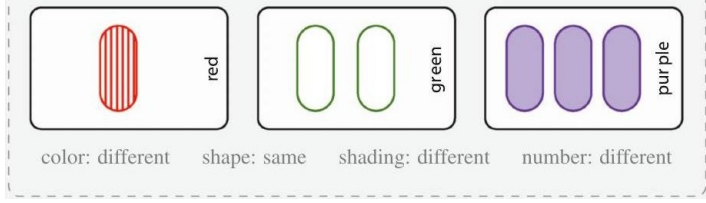
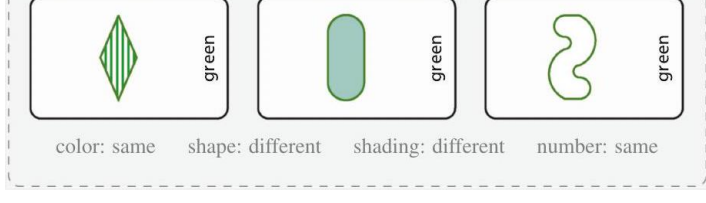
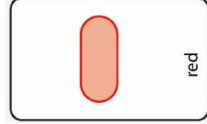
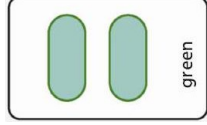
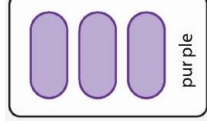
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra DIA_0056	118	—	—	—	
johnston-linear-matrix-algebra DIA_0057	118	—	—	—	
johnston-linear-matrix-algebra DIA_0058	118	—	—	—	
johnston-linear-matrix-algebra DIA_0059	123	—	—	—	
johnston-linear-matrix-algebra DIA_0060	124	—	—	—	
johnston-linear-matrix-algebra DIA_0061	127	—	—	—	

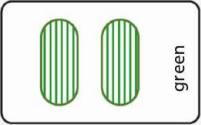
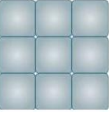
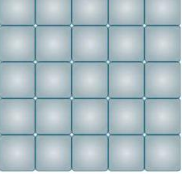
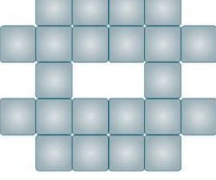
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-DIA_0062	133	—	—	—	
johnston-linear-matrix-algebra-DIA_0063	137	—	—	—	 (a) The line $y = x/2$ is a subspace, since lines through the origin are subspaces. (b) The line $y = x/2 + 1$ is not a subspace, since it does not go through the origin.
johnston-linear-matrix-algebra-DIA_0064	138	—	—	—	
johnston-linear-matrix-algebra-DIA_0065	145	—	—	—	
johnston-linear-matrix-algebra-DIA_0066	149	—	—	—	 (a) A linearly dependent set of 3 vectors that spans a plane in $\mathbb{R}^3$. (b) The set $\{(1, 2, 1), (2, 4, 2), (2, 1, 1)\}$ is linearly dependent, but removing $(2, 1, 1)$ from it changes its span from a plane to a line.

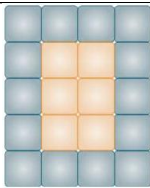
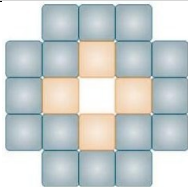
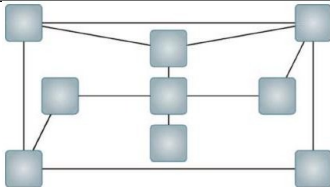
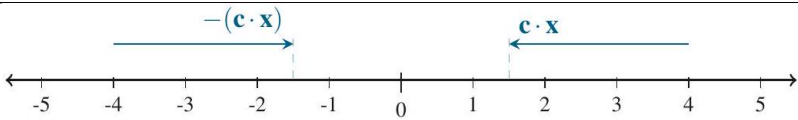
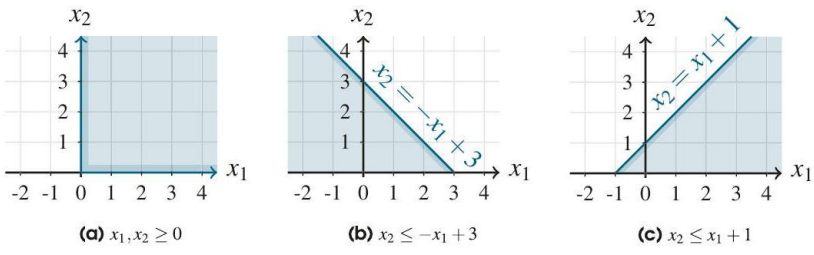
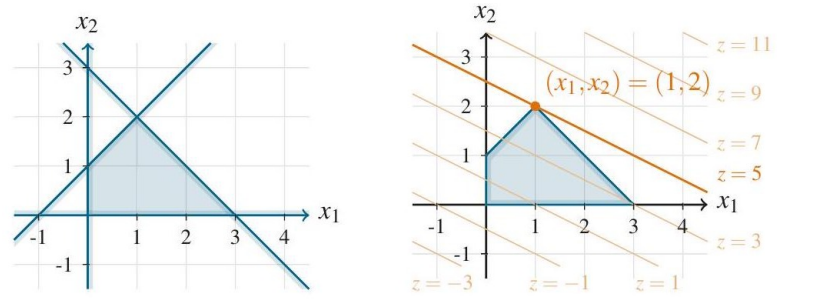
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra- DIA_0067	150	—	—	—	
johnston-linear-matrix-algebra- DIA_0068	155	—	—	—	
johnston-linear-matrix-algebra- DIA_0069	156	—	—	—	(a) A single vector spans a line. (b) Two non-parallel vectors span a plane. (c) Several vectors on a plane span that plane.
johnston-linear-matrix-algebra- DIA_0070	158	—	—	—	
johnston-linear-matrix-algebra- DIA_0071	159	—	—	—	(a) A basis of a line contains just 1 vector, so lines are 1-dimensional. (b) A basis of a plane contains 2 vectors, so planes are 2-dimensional.

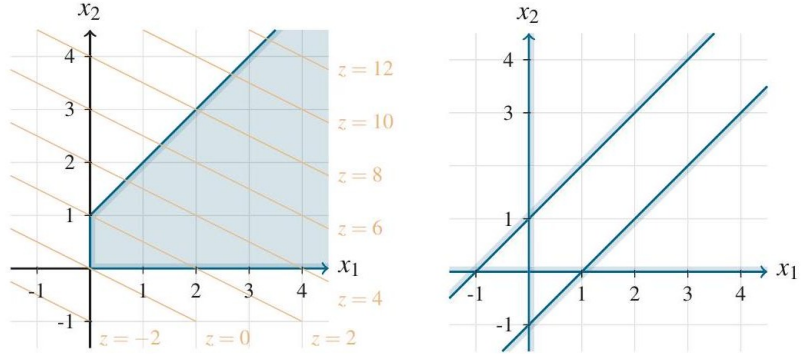
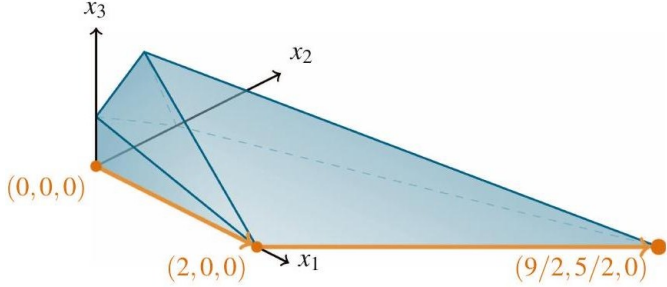
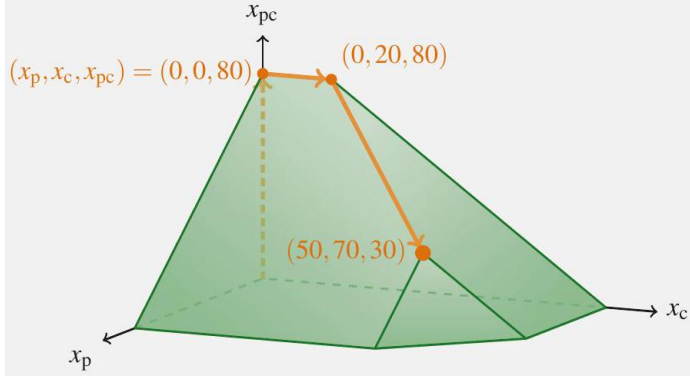
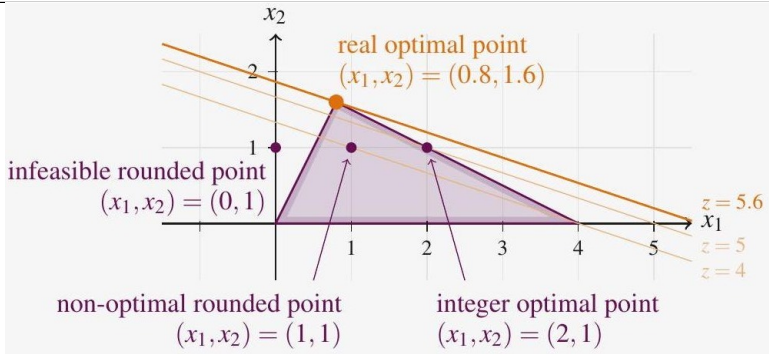
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-DIA_0072	161	—	—	—	
johnston-linear-matrix-algebra-DIA_0073	165	—	—	—	
johnston-linear-matrix-algebra-DIA_0074	166	—	—	—	
johnston-linear-matrix-algebra-DIA_0075	171	—	—	—	(a) A projection onto a line has rank 1, since its range is 1-dimensional. (b) A 3×3 matrix with rank 2 has a plane as its range.
johnston-linear-matrix-algebra-DIA_0076	174	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra DIA_0081	187	—	—	—	
johnston-linear-matrix-algebra DIA_0082	187	—	—	—	
johnston-linear-matrix-algebra DIA_0083	188	—	—	—	
johnston-linear-matrix-algebra DIA_0084	188	—	—	—	
johnston-linear-matrix-algebra DIA_0085	189	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-DIA_0086	191	—	—	—	 (a) In $\mathbb{Z}_3$, addition and multiplication work cyclically with the three values 0, 1, and 2. For (b) A clock illustrates the fact that $9 + 7 = 4$ in mod-12 arithmetic.
johnston-linear-matrix-algebra-DIA_0087	192	—	—	—	
johnston-linear-matrix-algebra-DIA_0088	192	—	—	—	
johnston-linear-matrix-algebra-DIA_0089	192	—	—	—	
johnston-linear-matrix-algebra-DIA_0090	194	—	—	—	
johnston-linear-matrix-algebra-DIA_0091	194	—	—	—	
johnston-linear-matrix-algebra-DIA_0092	194	—	—	—	
johnston-linear-matrix-algebra-DIA_0093	194	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-align- DIA_0094	194	—	—	—	
johnston-linear-matrix-align- DIA_0095	194	—	—	—	
johnston-linear-matrix-align- DIA_0096	194	—	—	—	
johnston-linear-matrix-align- DIA_0097	194	—	—	—	
johnston-linear-matrix-align- DIA_0098	194	—	—	—	
johnston-linear-matrix-align- DIA_0099	196	—	—	—	
johnston-linear-matrix-align- DIA_0100	196	—	—	—	
johnston-linear-matrix-align- DIA_0101	196	—	—	—	
johnston-linear-matrix-align- DIA_0102	196	—	—	—	
johnston-linear-matrix-align- DIA_0103	196	—	—	—	
johnston-linear-matrix-align- DIA_0104	196	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-DIA_0105	196	—	—	—	
johnston-linear-matrix-algebra-DIA_0106	196	—	—	—	
johnston-linear-matrix-algebra-DIA_0107	197	—	—	—	
johnston-linear-matrix-algebra-DIA_0108	198	—	—	—	
johnston-linear-matrix-algebra-DIA_0109	202	—	—	—	
johnston-linear-matrix-algebra-DIA_0110	202	—	—	—	 (a) The feasible region of a linear program is the intersection of the various half-planes defined by its constraints. (b) Of all the lines of the form $x_2 = -x_1/2 + z/2$, the one with largest value of z (i.e., largest y-intercept) that intersects the feasible region is the one with $z = 5$ (i.e., y-intercept $5/2$).

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-DIA_0111	203	—	—	—	 (a) Linear program (2.B.4) is unbounded. Its objective function gets larger as (x_1, x_2) moves to the top-right. (b) Linear program (2.B.5) is infeasible. There is no vector (x_1, x_2) that satisfies all four constraints simultaneously.
johnston-linear-matrix-algebra-DIA_0112	208	—	—	—	
johnston-linear-matrix-algebra-DIA_0113	210	—	—	—	
johnston-linear-matrix-algebra-DIA_0114	211	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-DIA_0115	213	—	—	—	
johnston-linear-matrix-algebra-DIA_0116	216	—	—	—	
johnston-linear-matrix-algebra-DIA_0117	218	—	—	—	
johnston-linear-matrix-algebra-DIA_0118	227	—	—	—	
johnston-linear-matrix-algebra-DIA_0119	228	—	—	—	
johnston-linear-matrix-algebra-DIA_0120	228	—	—	—	
johnston-linear-matrix-algebra-DIA_0121	228	—	—	—	
johnston-linear-matrix-algebra-DIA_0122	228	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-DIA_0123	228	—	—	—	
johnston-linear-matrix-algebra-DIA_0124	234	—	—	—	
johnston-linear-matrix-algebra-DIA_0125	249	—	—	—	
johnston-linear-matrix-algebra-DIA_0126	251	—	—	—	
johnston-linear-matrix-algebra-DIA_0127	253	—	—	—	
johnston-linear-matrix-algebra-DIA_0128	254	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-260 DIA_0129	259	—	—	—	$\mathbf{v}_1 = (2, 1, 0, 1, 2), \quad \mathbf{v}_2 = (1, 2, -1, 1, 0), \quad \mathbf{v}_3 = (0, 1, -3, 1, 1),$ $\mathbf{w}_1 = (1, -1, 1, 0, 2), \quad \mathbf{w}_2 = (1, 1, 2, 0, -1), \quad \mathbf{w}_3 = (2, 3, 1, 1, -1).$
johnston-linear-matrix-algebra-261 DIA_0130	261	—	—	—	$S \subseteq \mathbb{R}^n$ with $\dim(S) = k$
johnston-linear-matrix-algebra-263 DIA_0131	263	—	—	—	
johnston-linear-matrix-algebra-264 DIA_0132	264	—	—	—	

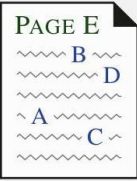
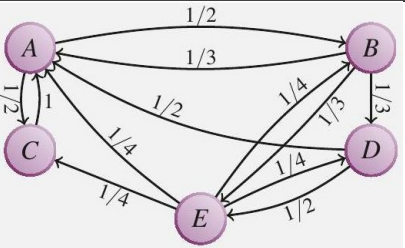
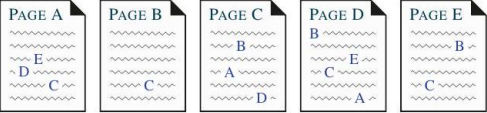
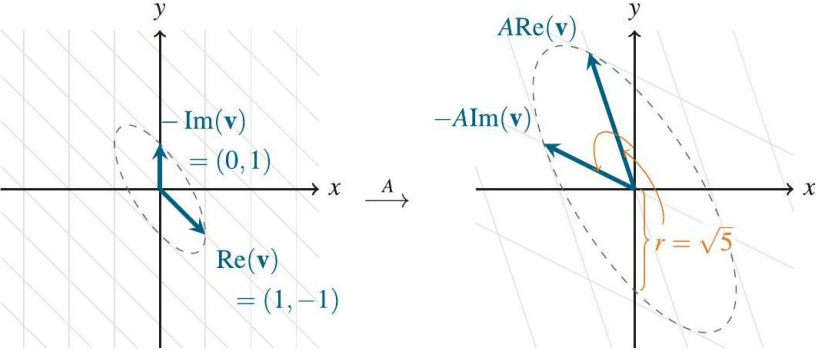
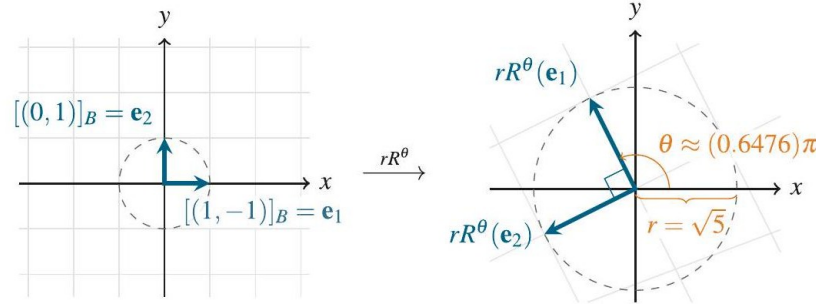
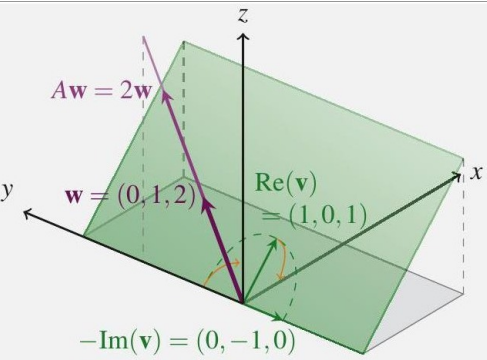
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-DIA_0133	264	—	—	—	
johnston-linear-matrix-algebra-DIA_0134	265	—	—	—	
johnston-linear-matrix-algebra-DIA_0135	272	—	—	—	
johnston-linear-matrix-algebra-DIA_0136	273	—	—	—	

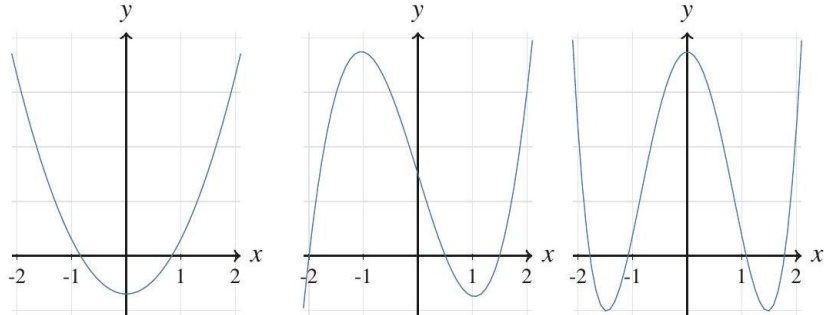
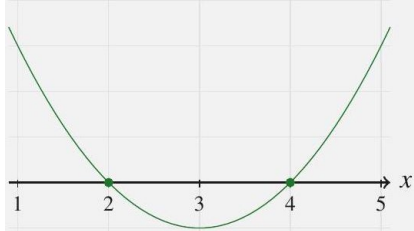
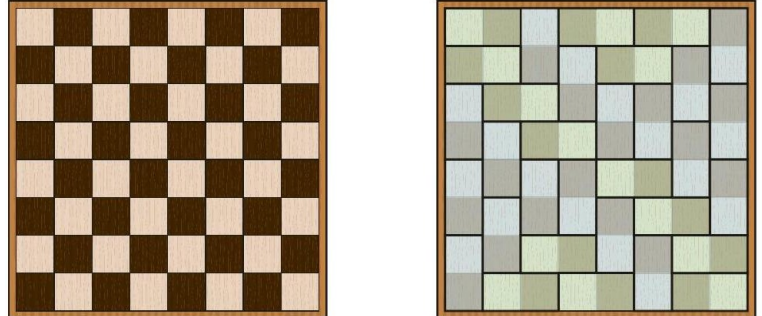
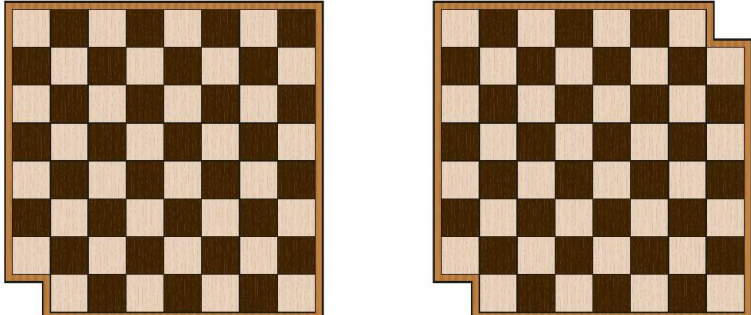
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-DIA_0137	275	—	—	—	
johnston-linear-matrix-algebra-DIA_0138	276	—	—	—	
johnston-linear-matrix-algebra-DIA_0139	277	—	—	—	
johnston-linear-matrix-algebra-DIA_0140	278	—	—	—	
johnston-linear-matrix-algebra-DIA_0141	278	—	—	—	
johnston-linear-matrix-algebra-DIA_0142	279	—	—	—	

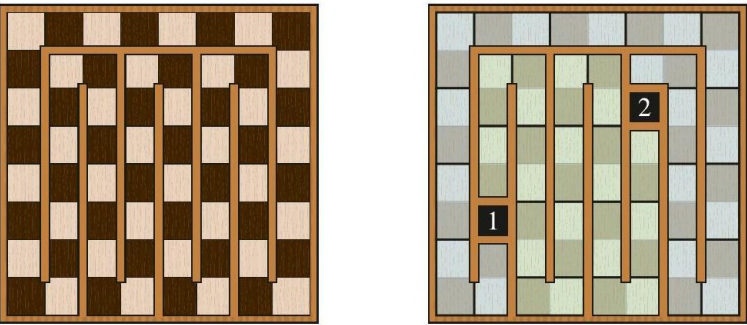
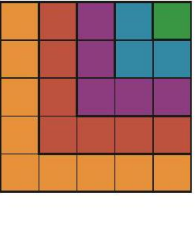
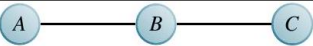
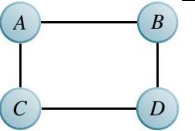
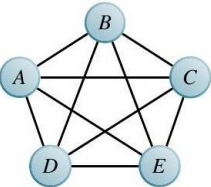
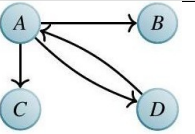
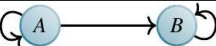
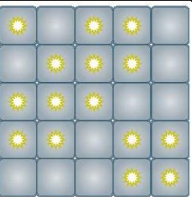
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-DIA_0143	280	—	—	—	
johnston-linear-matrix-algebra-DIA_0144	288	—	—	—	
johnston-linear-matrix-algebra-DIA_0145	289	—	—	—	
johnston-linear-matrix-algebra-DIA_0146	299	—	—	—	
johnston-linear-matrix-algebra-DIA_0147	303	—	—	—	
johnston-linear-matrix-algebra-DIA_0148	312	—	—	—	

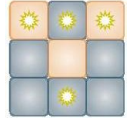
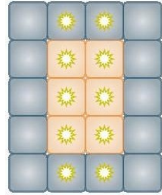
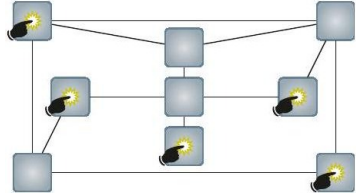
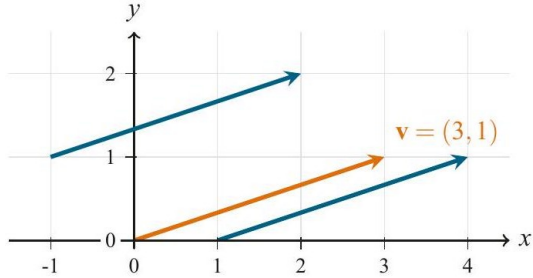
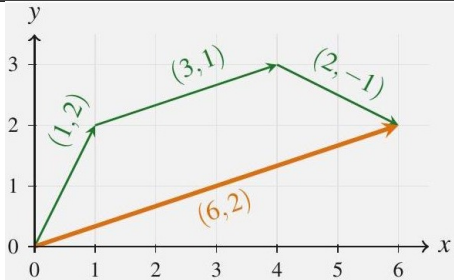
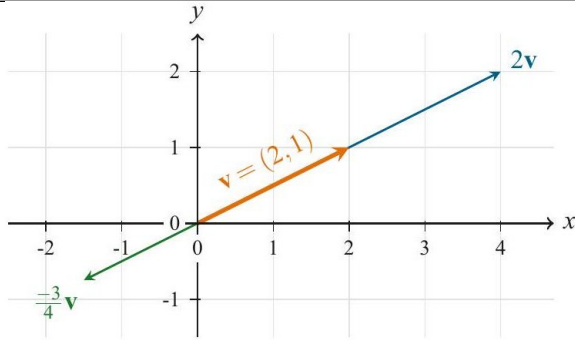
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-DIA_0149	317	—	—	—	
johnston-linear-matrix-algebra-DIA_0150	328	—	—	—	
johnston-linear-matrix-algebra-DIA_0151	331	—	—	—	
johnston-linear-matrix-algebra-DIA_0152	339	—	—	—	
johnston-linear-matrix-algebra-DIA_0153	344	—	—	—	
johnston-linear-matrix-algebra-DIA_0154	344	—	—	—	
johnston-linear-matrix-algebra-DIA_0155	344	—	—	—	
johnston-linear-matrix-algebra-DIA_0156	363	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-DIA_0157	371	—	—	—	
johnston-linear-matrix-algebra-DIA_0158	382	—	—	—	
johnston-linear-matrix-algebra-DIA_0159	386	—	—	—	
johnston-linear-matrix-algebra-DIA_0160	386	—	—	—	
johnston-linear-matrix-algebra-DIA_0161	386	—	—	—	
johnston-linear-matrix-algebra-DIA_0162	386	—	—	—	

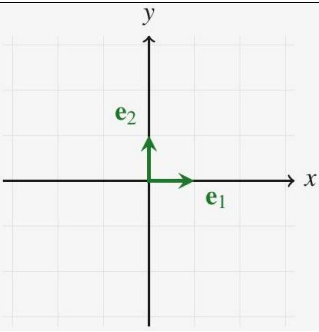
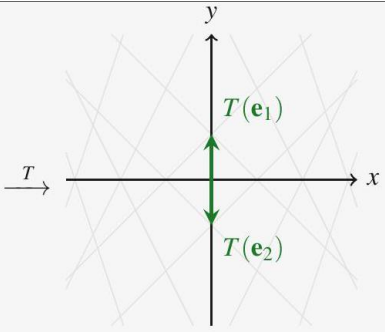
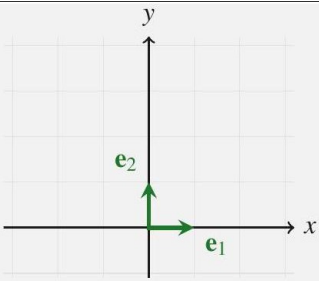
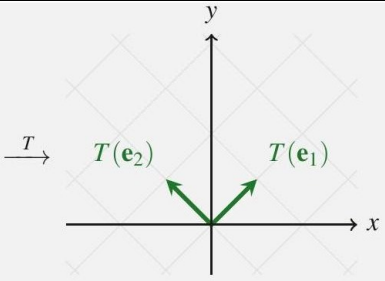
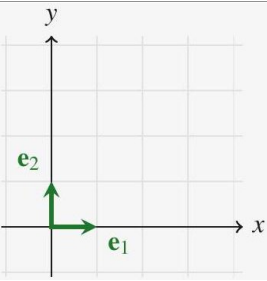
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-DIA_0163	386	—	—	—	
johnston-linear-matrix-algebra-DIA_0164	387	—	—	—	
johnston-linear-matrix-algebra-DIA_0165	390	—	—	—	
johnston-linear-matrix-algebra-DIA_0166	393	—	—	—	 (a) The matrix A rotates vectors around the edge of an ellipse and then stretches them by a factor of $r = \sqrt{5}$.  (b) In the basis $B = \{\text{Re}(\mathbf{v}), -\text{Im}(\mathbf{v})\} = \{(1, -1), (0, 1)\}$, the matrix A can be seen as rotating counter-clockwise by an angle of $\theta \approx (0.6476)\pi$ and stretching by a factor of $r = \sqrt{5}$.
johnston-linear-matrix-algebra-DIA_0167	398	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra- DIA_0168	422	—	—	—	 (a) A quadratic (degree 2). (b) A cubic (degree 3). (c) A quartic (degree 4).
johnston-linear-matrix-algebra- DIA_0169	423	—	—	—	
johnston-linear-matrix-algebra- DIA_0170	432	—	—	—	 (a) A standard 8×8 chessboard. (b) A domino tiling of a chessboard.
johnston-linear-matrix-algebra- DIA_0171	432	—	—	—	 (a) Any chessboard with one square removed cannot be tiled by dominoes. (b) This board with two opposite corners removed cannot be tiled by dominoes.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-atlas- DIA_0172	433	—	—	—	 (a) An arrangement of "walls" on a chess-board that will help us tile it. (b) A tiling of a board with (opposite-colored) squares "1" and "2" removed.
johnston-linear-matrix-atlas- DIA_0173	436	—	—	—	
johnston-linear-matrix-atlas- DIA_0174	454	—	—	—	
johnston-linear-matrix-atlas- DIA_0175	454	—	—	—	
johnston-linear-matrix-atlas- DIA_0176	455	—	—	—	
johnston-linear-matrix-atlas- DIA_0177	455	—	—	—	
johnston-linear-matrix-atlas- DIA_0178	455	—	—	—	
johnston-linear-matrix-atlas- DIA_0179	467	—	—	—	
johnston-linear-matrix-atlas- DIA_0180	467	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_DIA_0181	468	—	—	—	
johnston-linear-matrix-algebra_DIA_0182	468	—	—	—	
johnston-linear-matrix-algebra_DIA_0183	468	—	—	—	$A = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra_DIA_0184	468	—	—	—	
johnston-linear-matrix-algebra_PIC_0001	17	—	—	—	
johnston-linear-matrix-algebra_PIC_0002	19	—	—	—	
johnston-linear-matrix-algebra_PIC_0003	20	—	—	—	

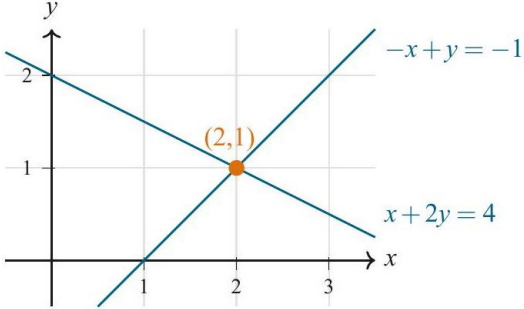
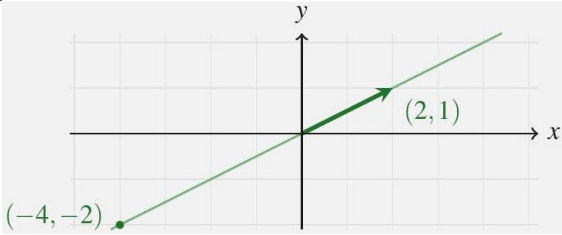
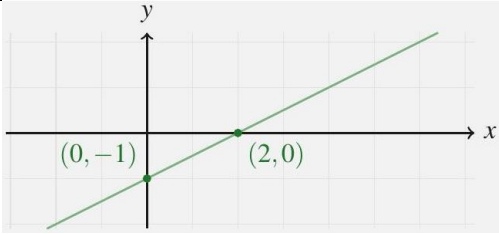
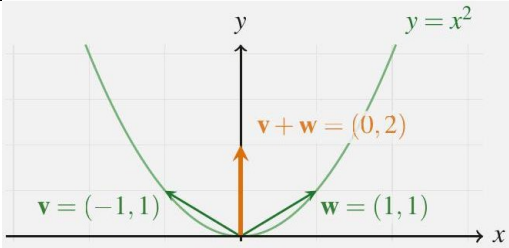
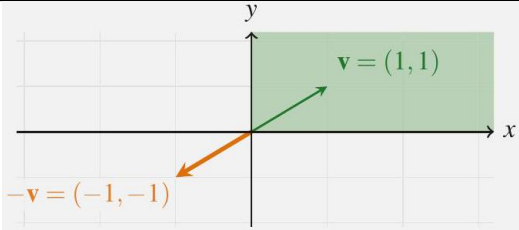
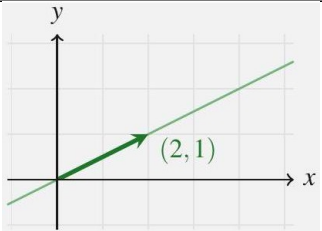
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_PIC_0004	21	—	—	—	
johnston-linear-matrix-algebra_PIC_0005	22	—	—	—	
johnston-linear-matrix-algebra_PIC_0006	26	—	—	—	
johnston-linear-matrix-algebra_PIC_0007	31	—	—	—	
johnston-linear-matrix-algebra_PIC_0008	52	—	—	—	
johnston-linear-matrix-algebra_PIC_0009	52	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_PIC_0010	52	—	—	—	
johnston-linear-matrix-algebra_PIC_0011	52	—	—	—	
johnston-linear-matrix-algebra_PIC_0012	53	—	—	—	
johnston-linear-matrix-algebra_PIC_0013	53	—	—	—	
johnston-linear-matrix-algebra_PIC_0014	55	—	—	—	

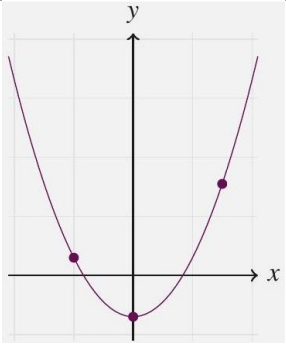
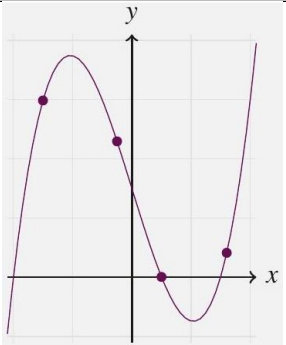
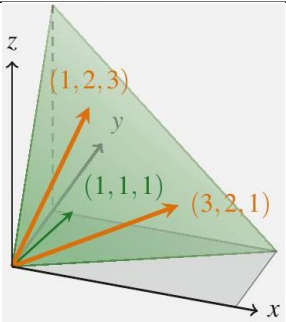
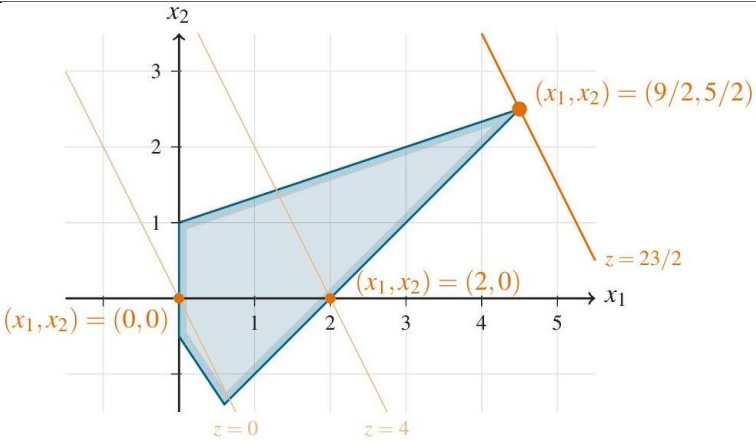
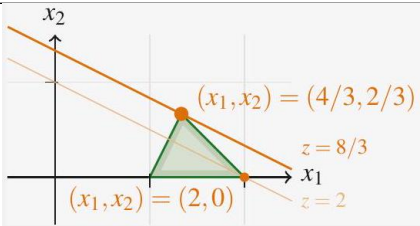
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_PIC_0015	55	—	—	—	
johnston-linear-matrix-algebra_PIC_0016	56	—	—	—	
johnston-linear-matrix-algebra_PIC_0017	56	—	—	—	
johnston-linear-matrix-algebra_PIC_0018	60	—	—	—	
johnston-linear-matrix-algebra_PIC_0019	60	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_PIC_0020	62	—	—	—	
johnston-linear-matrix-algebra_PIC_0021	64	—	—	—	
johnston-linear-matrix-algebra_PIC_0022	64	—	—	—	
johnston-linear-matrix-algebra_PIC_0023	68	—	—	—	
johnston-linear-matrix-algebra_PIC_0024	68	—	—	—	
johnston-linear-matrix-algebra_PIC_0025	68	—	—	—	
johnston-linear-matrix-algebra_PIC_0026	68	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_PIC_0027	68	—	—	—	
johnston-linear-matrix-algebra_PIC_0028	68	—	—	—	
johnston-linear-matrix-algebra_PIC_0029	72	—	—	—	
johnston-linear-matrix-algebra_PIC_0030	77	—	—	—	
johnston-linear-matrix-algebra_PIC_0031	79	—	—	—	
johnston-linear-matrix-algebra_PIC_0032	79	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-PIC_0033	94	—	—	—	
johnston-linear-matrix-algebra-PIC_0034	137	—	—	—	
johnston-linear-matrix-algebra-PIC_0035	137	—	—	—	
johnston-linear-matrix-algebra-PIC_0036	138	—	—	—	
johnston-linear-matrix-algebra-PIC_0037	139	—	—	—	
johnston-linear-matrix-algebra-PIC_0038	140	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-PIC_0039	140	—	—	—	
johnston-linear-matrix-algebra-PIC_0040	141	—	—	—	
johnston-linear-matrix-algebra-PIC_0041	142	—	—	—	
johnston-linear-matrix-algebra-PIC_0042	143	—	—	—	
johnston-linear-matrix-algebra-PIC_0043	149	—	—	—	
johnston-linear-matrix-algebra-PIC_0044	152	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra- PIC_0045	152	—	—	—	
johnston-linear-matrix-algebra- PIC_0046	152	—	—	—	
johnston-linear-matrix-algebra- PIC_0047	160	—	—	—	
johnston-linear-matrix-algebra- PIC_0048	208	—	—	—	
johnston-linear-matrix-algebra- PIC_0049	214	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-PIC_0050	239	—	—	—	
johnston-linear-matrix-algebra-PIC_0051	240	—	—	—	
johnston-linear-matrix-algebra-PIC_0052	249	—	—	—	
johnston-linear-matrix-algebra-PIC_0053	250	—	—	—	
johnston-linear-matrix-algebra-PIC_0054	253	—	—	—	
johnston-linear-matrix-algebra-PIC_0055	256	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-PIC_0056	257	—	—	—	
johnston-linear-matrix-algebra-PIC_0057	258	—	—	—	
johnston-linear-matrix-algebra-PIC_0058	259	—	—	—	$\mathbf{w}_1 = (1, -1, 1, 0, 2), \quad \mathbf{w}_2 = (1, 1, 2, 0, -1), \quad \mathbf{w}_3 = (2, 3, 1, 1, -1).$
johnston-linear-matrix-algebra-PIC_0059	274	—	—	—	
johnston-linear-matrix-algebra-PIC_0060	274	—	—	—	
johnston-linear-matrix-algebra-PIC_0061	280	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-PIC_0062	281	—	—	—	
johnston-linear-matrix-algebra-PIC_0063	300	—	—	—	
johnston-linear-matrix-algebra-PIC_0064	301	—	—	—	
johnston-linear-matrix-algebra-PIC_0065	301	—	—	—	
johnston-linear-matrix-algebra-PIC_0066	304	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra- PIC_0067	327	—	—	—	
johnston-linear-matrix-algebra- PIC_0068	331	—	—	—	
johnston-linear-matrix-algebra- PIC_0069	331	—	—	—	
johnston-linear-matrix-algebra- PIC_0070	352	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra- PIC_0071	372	—	—	—	
johnston-linear-matrix-algebra- PIC_0072	376	—	—	—	
johnston-linear-matrix-algebra- PIC_0073	379	—	—	—	
johnston-linear-matrix-algebra- PIC_0074	392	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra- PIC_0075	414	—	—	—	
johnston-linear-matrix-algebra- PIC_0076	418	—	—	—	
johnston-linear-matrix-algebra- PIC_0077	418	—	—	—	
johnston-linear-matrix-algebra- PIC_0078	420	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-PIC_0079	421	—	—	—	
johnston-linear-matrix-algebra-PIC_0080	421	—	—	—	
johnston-linear-matrix-algebra-PIC_0081	423	—	—	—	
johnston-linear-matrix-algebra-PIC_0082	424	—	—	—	
johnston-linear-matrix-algebra-PIC_0083	446	—	—	—	
johnston-linear-matrix-algebra-PIC_0084	446	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-atlas- PIC_0085	446	—	—	—	
johnston-linear-matrix-atlas- PIC_0086	446	—	—	—	
johnston-linear-matrix-atlas- PIC_0087	446	—	—	—	
johnston-linear-matrix-atlas- PIC_0088	446	—	—	—	
johnston-linear-matrix-atlas- PIC_0089	452	—	—	—	
johnston-linear-matrix-atlas- PIC_0090	452	—	—	—	